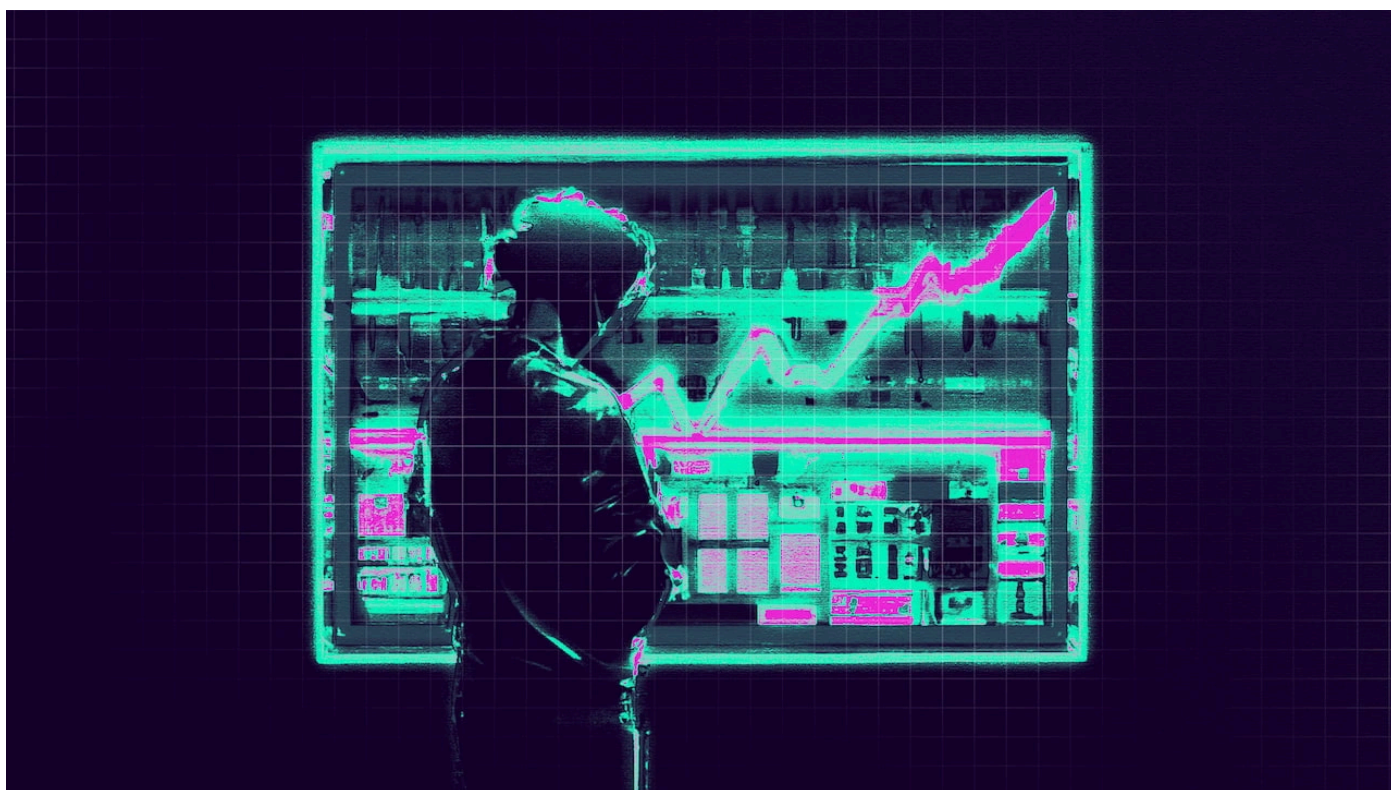


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7 min read

What is QPS Optimization?

Whenever you visit a website to read the latest news, you will see an ad. Let's say you've looked for a new car recently and see an ad for the new BMW i4. Is this magic? How did it happen?



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Well, with a ton of complex infrastructure - that is the short answer.

Each ad you see on a website with an optimized adstack is probably going through several actors in the ad ecosystem in pursuit of higher yield, personalized ads, and good conversions. Once you open the browser, requests may go through different platforms, such as the open source Prebid.org, centralized Amazon A9, and ad-serving built-in software - Open Bidding by Google. This infrastructure passes queries to each SSP, or Supply Side Platform, connected to it.

See, before an ad is served to you, there are several phases - a publisher has the inventory, then there are platforms that connect it to players that consolidate many publishers' inventory - the SSPs. Usually, a website works with 10 to 30 different SSPs, and each must receive the information that there is a free space to show their ad.

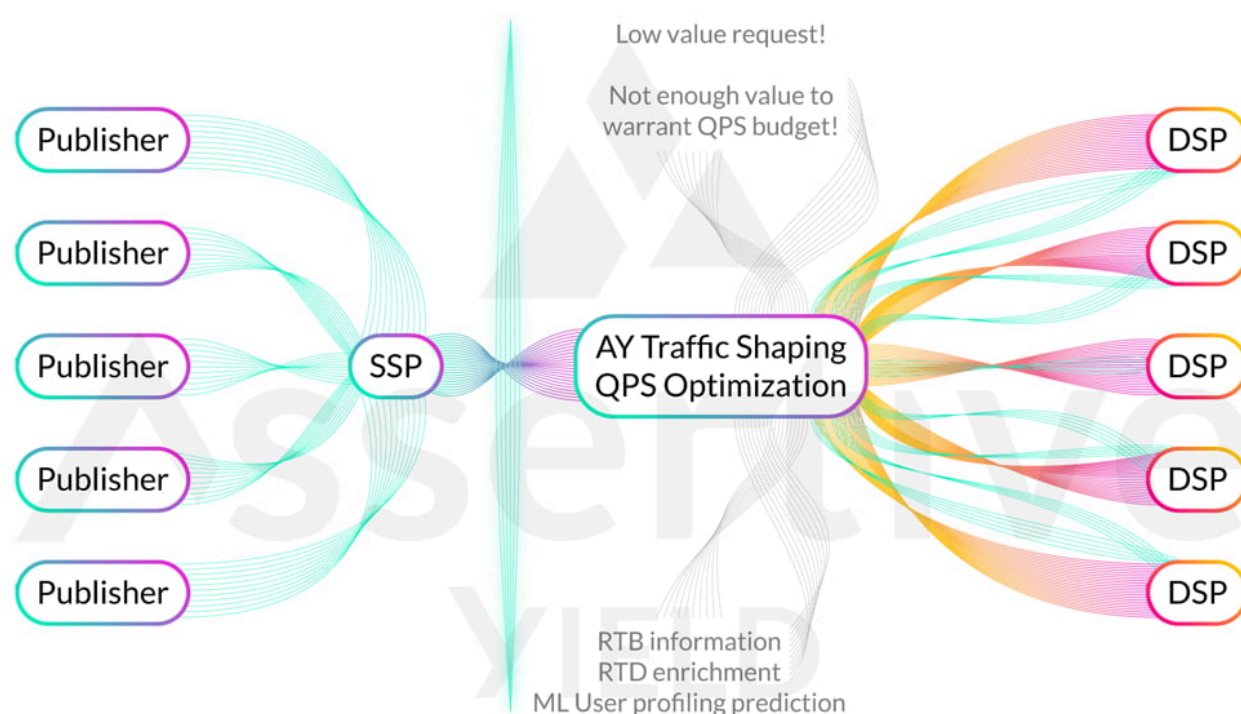


Image shows the supply path and how a traffic shaping solution ensures green traffic and lower costs.

Then the SSPs pass this information to the DSPs (Demand Side Platforms). The DSPs consolidate advertisers and develop the technology that enables them to spend their ad budget optimally in the right place at the right time.

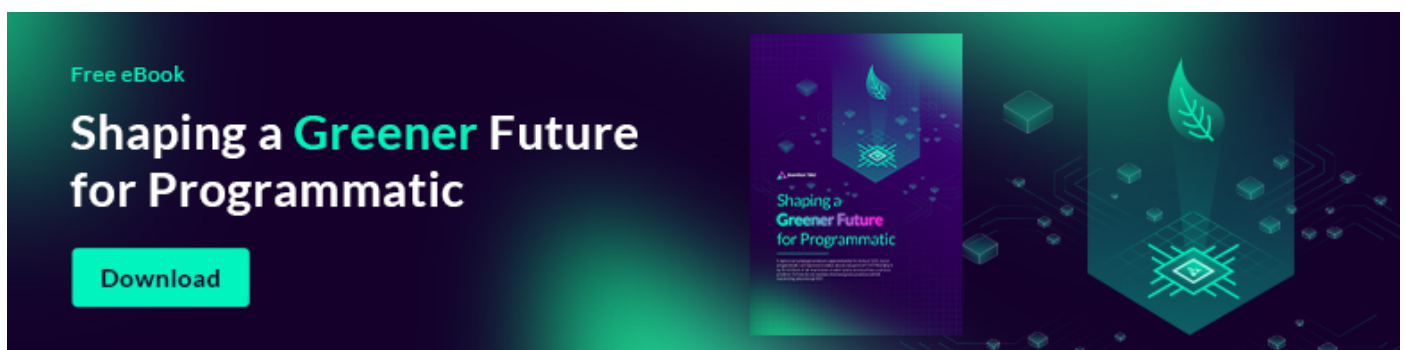
In theory, an SSP like Appnexus sends a request to a DSP like The Trade Desk, there is a mix of cookie-matching in the process of delivery rates, ad sizes, cappings, etc., and in the end - The Trade Desk sees an opportunity for its client BMW, and sends back information that it can give as much as 1 dollar per 1000 impressions like these. Each so-called bid is then transferred to the Ad Server (let's say Google

Ad Manager), and as it collects them, it checks which is the highest price and displays the advertisers' banner ad from the DSP.

It's a rather complex, technologically challenging infrastructure, which takes seconds to display a hopefully relevant ad.

And in this competitive sphere, Publishers are adding more and more SSPs to share their inventory with, SSPs are adding more and more publishers to get more commissions from the inventory, and the DSPs are adding more and more SSPs to make sure they can find the best ad recipients and prices there.

As you can imagine, all these components need to communicate with each other resulting in trillions of requests between each SSP and DSP every month. Lots of service resources, high traffic costs, and [huge amounts of carbon emissions.](#)



What is QPS anyway?

What is Query per second(QPS) Optimization?

What does QPS mean?

QPS is the number of information requests a service handles per second. Each one of the ad spots on the site usually has its own requests flying around to different SSPs, which then multiply them by the number of DSPs they are working with. Often resulting in hundreds of requests for a single ad placement. . And the truth is - there is a lot of “spam” in these requests, which most DSPs won't buy.

Therefore, many DSPs enforce strict QPS limitations, meaning they provide each of their supply partners with a set budget of how many requests they may send per second.

Therefore, optimizing the QPS is vital to gain optimal results from the available budget. This technique has transformed common platforms into online giants as we know them today. Only through a sophisticated process can you show the right Ad to the right person at the right time.

What is the role played by QPS in this regard?

How does QPS Optimization work?

Real-time bidding technologies have seen a meteoric rise, especially since the pandemic. However, traffic also increased when people got more time staying home. This was stressful and challenging for the ad industry. It was evident when ad spending decreased. And now, with businesses going online worldwide, the traffic between the SSP and DSP has become a bottleneck, big time!

SSP players need more infrastructure to accommodate the multitude of requests coming in from publishers. This surge of traffic requires them to invest more in Cloud resources and increase expenses.

Next, coming to the receiving end.

DSPs get bombarded with requests from the supply-side, increasing their cloud costs without a change in revenue. This issue ultimately affects the partnerships they maintain with advertisers and agencies.

Some partnerships stipulate limited SSP queries to avoid network overload, making things even more challenging. For instance, 5,000 queries per second are all they are allowed to transmit to the DSP. The subsequent ones are ignored, leaving a potential piece of the pie on the table because there is no one to eat it.

But what if I tell you that you can eat your pie and still have it because there is a solution?

Now you know why optimizing the queries per second is pivotal for dynamic Ad campaigns and their revenue generation capability. So what is the solution, then?

Discover how GumGum achieved a 30% QPS on average reduction using Traffic Shaping. Read the full story



QPS Optimization Solutions

We present our unique solution that enables SSPs to scale unprecedented revenue via QPS Optimization powered by machine learning - [AY Traffic Shaping](#).

Unclutter the Demand-Supply bottlenecks and deliver more quality and quantity of online ads like never before!

AY Traffic Shaping makes this happen. How?

![4 colleagues high-fiving ay traffic shaping product]("High-fiving AY Traffic Shaping Product")

How does AY Traffic Shaping work?

We combine the triple benefits of sophisticated Machine Learning algorithms, Data Enrichment, and User Profiling to proactively identify the most valuable traffic for each DSP. How does this help?

- Up to 40% cost savings due to optimized QPS and smoother infrastructure utilization
- 6-12% increase in fill rates due to better quality of traffic
- Increased bid rates
- Solidified relationships with your DSP partners
- Eliminate up to 60% of non-responsive bid requests
- [Shape greener traffic for Programmatic](#)

ing using AY's Artificial Intelligence component

When an SSP integrates [AY Traffic Shaping](#), it automatically picks the highest value requests for each DSP within the QPS limit and further reduces requests which are pure spam. Oftentimes reductions of +50% are seen while losing less than 0.5% of bids and increasing overall revenue by 5-10%.

Adtech Solution for Traffic Shaping

Assertive Yield has been the go-to partner for publishers across industries. Our unmatched brand reputation has been made possible due to our relentless efforts to help them manage their demands better, leading to increased revenues. Now we are bringing our machine learning optimization from the publisher side to the SSPs.

![[two male colleagues giving thumbs-up with both hands to Assertive Yield]]("Thumbs-Up Assertive Yield")

Conclusion

Assertive Yield is a company with a strong record of helping publishers achieve better results by optimizing their demand. In pursuit of better bid rates and fewer client-to-server requests in the browsers, we can predict the probability of an SSP returning a positive request before it reaches the servers. And now, we do the same with SSPs, enabling them to decrease queries to DSPs.

Reach Higher profitability with lower cost with AY Traffic Shaping. Shape and optimize your traffic to generate more profitability from your DSP partners, using assertive user profiling to reach more profitable matchmaking.

[Reach out to us](#) today for more info about our revolutionary Traffic Shaping Solution.

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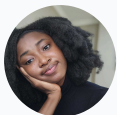
Explore more about

Efficiency & Sustainability

Check out the associated AY solution

Traffic Shaping

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Angela is a wordsmith who skillfully transforms complex concepts into engaging content, making significant contributions to the advertising industry. Leveraging her wealth of expertise as a content writer, she crafts compelling narratives that seamlessly bridge technology and communication, whether with her pen or keyboard.





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Q2'25 Publisher Trends

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Frequently asked questions

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What is QPS Optimization?



How does QPS optimization work?



What are the best QPS optimization solutions?





DPO Mastery: Traffic Shaping for Revenue Growth

This exclusive four-episode series brings together industry experts Chris Kane, Nils Lind, and Rob Beeler to tackle the biggest challenges and opportunities for Digital Publishers, SSPs, and DSPs.

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